

CHAPTER 3. DIFFERENTIATION

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3.1 An Introduction to the Derivative: Tangent lines

The **tangent line** to the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is the line through P with slope

$$m = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (1)$$

provided that this limit exists. Alternatively

$$m = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad (2)$$

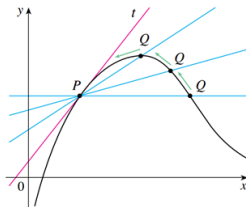


Figure: Slope of a curve

Example 1

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Recall: **Point–slope form** for a line through the point (x_0, y_0) with slope m :

$$y - y_0 = m(x - x_0) \quad (3)$$

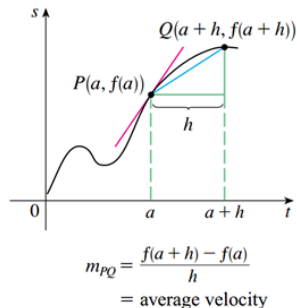
3.1 An Introduction to the Derivative: Velocities

Suppose an object moves along a straight line according to an equation of motion

$$s = f(t)$$

The **average velocity** over the time interval $[a, a + h]$ is

$$\begin{aligned}\text{average velocity} &= \frac{\text{displacement}}{\text{time}} \\ &= \frac{f(a + h) - f(a)}{h}\end{aligned}$$



The **velocity** (or **instantaneous velocity**) $v(a)$ at time a is

$$v(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \quad (4)$$

Example 2

If a rock is thrown upward on the planet Mars with a velocity of 10 m/s, its height in meters t seconds later is given by $y = 10t - 1.86t^2$.

(a) Find the average velocity over the given time intervals:

$$[1, 2], \quad [1, 1.5], \quad [1, 1.1], \quad [1, 1.01], \quad [1, 1.001].$$

(b) Estimate the instantaneous velocity when $t = 1$.

3.1 An Introduction to Derivatives: Derivatives

Definition 3 (Derivative)

The **derivative of a function f at a number a** , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Alternatively, we can write

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Example 4

Find the derivative of the function $f(x) = x^2 - 3x + 1$ at the number a ?

Example 5

Find an equation of the tangent line to the parabola $y = x^2 - 3x + 1$ at the point $P(1, -1)$.

Example 6

Find an equation of the normal line to the parabola $y = x^2 - 3x + 1$ at the point $P(1, -1)$.

Normal line to a graph: The normal line to a graph of f at the point P is the line that is perpendicular to the tangent line at the point P .

Derivative Notation

Given the function $y = f(x)$. Then the function f' defined through

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (5)$$

is so-called the **derivative of f** .

LEIBNIZ NOTATION:

- Derivative of a function f

$$\frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = f'(x)$$

- Derivative of a function f at a specific number a

$$\left. \frac{dy}{dx} \right|_{x=a} = \left. \frac{df}{dx} \right|_{x=a} = f'(a)$$

Example 7

(a) Differentiate $f(x) = \sqrt{x}$.

(b) Find $\left. \frac{dy}{dx} \right|_{x=1}$ if $y = x^2$.

Definition 8 (Differentiability)

A function f is **differentiable at** a if $f'(a)$ exists. It is **differentiable on an open interval** (a, b) (or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$) if it is differentiable at every number in the interval.

Example 9

Show that the absolute value function $f(x) = |x|$ is not differentiable at 0?

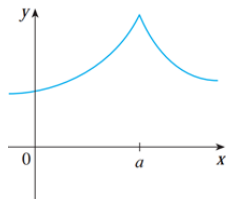
NOTICE:

- Differentiability $\implies$ continuity, that is,

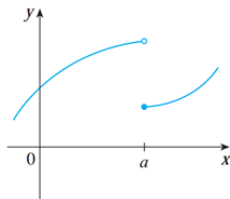
If f is differentiable at a , then f is continuous at a .

- The converse is false.

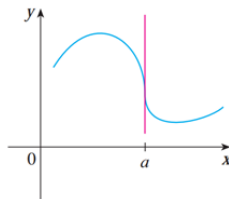
HOW CAN A FUNCTION FAIL TO BE DIFFERENTIABLE?



(a) A corner



(b) A discontinuity



(c) A vertical tangent

Figure: Common examples where a derivative does not exist

3.2 Techniques of Differentiation

Derivatives of Polynomials, Exponential and Logarithmic Functions

1. Derivative of a Constant Function

$$\frac{d}{dx}(c) = 0$$

2. The Power Rule: If n is any real number, then

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

3. Derivative of the Exponential and Logarithmic Functions

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(a^x) = a^x \ln a, \quad (0 < a \neq 1)$$

$$\frac{d}{dx}(\ln |x|) = \frac{1}{x}$$

$$\frac{d}{dx}(\log_a |x|) = \frac{1}{x \ln a}$$

3.2 Techniques of Differentiation

General Formulas

If f and g are both differentiable, then

(1) $[cf(x)]' = cf'(x)$, where c is a constant.

(2) The sum and the difference rules:

$$[f(x) \pm g(x)]' = f'(x) \pm g'(x)$$

(3) The product rule:

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

(4) The quotient rule:

$$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

3.2 Techniques of Differentiation

Higher Derivatives

Let f be a differentiable function.

- The **first derivative** of f

$$y' = f'(x) = \frac{dy}{dx} = \frac{d}{dx}f(x)$$

- The **second derivative** of f

$$y'' = f''(x) = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}$$

- The **n th derivative** of f is denoted by

$$y^{(n)} = f^{(n)}(x) = \frac{d}{dx} \left(\frac{d^{n-1}y}{dx^{n-1}} \right) = \frac{d^ny}{dx^n}$$

Example 10

If $f(x) = x^3 - x$, find $f'''(x)$ and $f^{(4)}(x)$.

3.3 Derivatives of Trigonometric Functions

Derivatives of Trigonometric Functions	The Chain Rule
$[\sin x]' = \cos x$	$[\sin u]' = u' \cos u$
$[\cos x]' = -\sin x$	$[\cos u]' = -u' \sin u$
$[\tan x]' = 1 + \tan^2 x = \frac{1}{\cos^2 x}$	$[\tan u]' = u' (1 + \tan^2 u) = \frac{u'}{\cos^2 u}$
$[\cot x]' = -(1 + \cot^2 x) = -\frac{1}{\sin^2 x}$	$[\cot u]' = -u' (1 + \cot^2 u) = -\frac{u'}{\sin^2 u}$
$[\csc x]' = -\csc x \cot x$	$[\csc u]' = -u' \csc u \cot u$
$[\sec x]' = \sec x \tan x$	$[\sec u]' = u' \sec u \tan u$

Example 11

Find the standard form equation for the line tangent to the graph of

$$f(x) = \frac{3x^2 + 5}{2x^2 + x - 3}$$

at the point where $x = -1$.

Example 12

Let $y = (x - 2)(x^2 + 4x - 7)$. Find all points on this curve where the tangent line is horizontal.

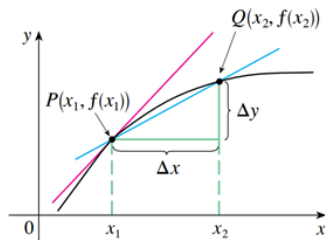
3.4 Rate of Changes and Applications

The **average rate of change of y w.r.t to x** over the interval $[x_1, x_2]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

where

- $\Delta x = x_2 - x_1$ is the **increment** of x
- Δy is the **increment** of y



average rate of change = m_{PQ}

instantaneous rate of change =
slope of tangent at P

The **(instantaneous) rate of change of y with respect to x** at $x = x_1$ is defined by

$$\text{rate of change at } x_1 = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(x_1)$$

Example 13 (Exercise 38, p. 171)

A bucket containing 5 gal of water has a leak. After t second, there are

$$Q(t) = 5 \left(1 - \frac{t}{25}\right)^2$$

gallons of water in the bucket.

- At what rate (to the nearest hundredth gal) is water leaking from the bucket after 2 seconds?
- How long does it take for all the water to leak out of the bucket?
- At what rate is the water leaking when the last drop leaks out?

3.4 Rate of Changes and Applications

Rectilinear Motion

Let $s = f(t)$ be the position function of a particle that is moving in a straight line, then we have that:

- The instantaneous **velocity** at the time t is

$$v(t) = s'(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}.$$

- The **acceleration** at the time t is

$$a(t) = v'(t) = s''(t)$$

Discuss the motion of a particle

Velocity of the particle:

- the particle is **advancing** if $v(t) > 0$;
- it is **retreating** if $v(t) < 0$;
- it is **stationary** if $v(t) = 0$.

Acceleration of the particle:

- the particle is **accelerating** when $a(t) > 0$;
- the particle is **decelerating** when $a(t) < 0$.

The speed of the particle $|v(t)|$:

- The speed *increases* when the velocity and acceleration have the same signs.
- The speed *decreases* when the velocity and acceleration have the opposite signs.

Example 14

The position of a particle is given by the equation

$$s(t) = t^3 - 6t^2 + 9t$$

where t is measured in seconds and s in meters. Find the initial position, velocity, acceleration for the particle and discuss the motion. Compute the total distance traveled by the particle during the first five seconds.

Example 15

If a ball is thrown vertically upward with a velocity of 80 ft/s, then its height after t seconds is $s(t) = 80t - 16t^2$.

- (a) What is the maximum height reached by the ball?
- (b) What is the velocity of the ball when it is 96 ft above the ground on its way up? On its way down?

3.4 Rate of Changes and Applications

Falling Body Problems

Falling Body Problems

Assume that an object is projected (that is, thrown, fired, dropped, etc.) vertically in such a way that the only acceleration acting on the object is the constant downward accelerating g due to gravity.

Then at the time t the height of the projectile is given by the following formula

$$h(t) = -\frac{1}{2}gt^2 + v_0t + s_0 \quad (6)$$

where

- s_0 is the **initial height**
- v_0 is the **initial velocity**
- g is the **acceleration** due to gravity, $g \approx 9.8m/s^2$ (or $32ft/s^2$).

Example 16

Suppose a person standing at the top of the Tower of Pisa (176 ft high) throws a ball directly upward with an initial speed 96 ft/s.

- Find the ball's height, its velocity, and its acceleration at the time t .
- When does the ball hit the ground and is its impact velocity?
- How far does the ball travel during its flight?

Example 17 (Exercise 39, p. 171)

A person standing at the edge of the cliff throws a rock directly upward. It is observed that 2 seconds later the rock is at its maximum height (in ft) and that 5 seconds after that, it hits the ground at the base of the cliff.

- What is the initial velocity of the rock?
- How high is the cliff?
- What is the velocity of the rock at time t ?
- With what velocity does the rock hit the ground?

3.5 The Chain Rule

The Chain Rule

If g is differentiable at x and f is differentiable at $f(g(x))$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F is given by the product

$$F'(x) = f'(g(x)) g'(x)$$

The Chain Rule can be written either in the prime notation

$$(f \circ g)'(x) = f'(g(x)) g'(x)$$

or, if $y = f(u)$ and $u = g(x)$, in Leibniz notation:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Example 18

Find $F'(x)$, if $F(x) = \sqrt{x^2 + 1}$.

Example 19

Differentiate the functions

a. $y = (x^3 - 1)^{100}$ **b.** $y = e^{\sin x}$ **c.** $g(t) = \left(\frac{t-1}{2t+1} \right)^9$

Example 20 (Exercise 47, p. 179)

An importer of Rwandan coffee estimates that local consumers will buy approximately

$$D(p) = \frac{4,374}{p^2}$$

pound of coffee per week when the price is p dollars per pound. It is estimated that t weeks from now, the price of Rwandan coffee will be

$$p(t) = 0.02t^2 + 0.01t + 6$$

dollar per pound. At what rate will the weekly demand for the coffee be changing with respect to time 10 weeks from now? Will the demand be increasing or decreasing?

3.6 The Implicit Differentiation

In this subsection

- 1 What is implicit function?
- 2 How to differentiate an implicit function?
- 3 Applications: Differentiate inverse functions, logarithmic differentiation.

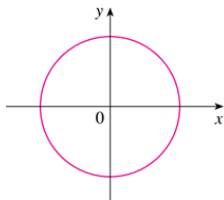
The functions that can be described by expressing one variable explicitly in terms of another variable which is so-called **explicit functions**

$$y = \sqrt{x^3 + 1} \quad \text{or} \quad y = x \sin x$$

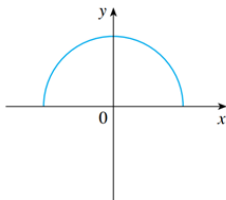
or, in general, $y = f(x)$.

Some functions $y = y(x)$ are defined implicitly through an equation

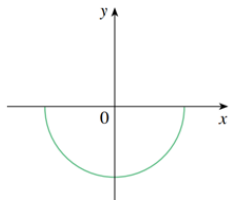
$$x^2 + y^2 = 25 \quad \text{or} \quad x^3 + y^3 = 6xy$$



(a) $x^2 + y^2 = 25$



(b) $f(x) = \sqrt{25 - x^2}$



(c) $g(x) = -\sqrt{25 - x^2}$

Implicit Differentiation

Suppose $y = y(x)$ is an implicit function defined through an equation

$$F(x, y) = 0$$

To find the derivative dy/dx :

- **Step 1:** Differentiate both sides of the equation with respect to x .
- **Step 2:** Solve the resulting equation for dy/dx .

Example 21

(a) If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $P(3, 4)$.

Example 22

- (a) Find y' if $x^3 + y^3 = 6xy$.
- (b) Find the tangent to the folium of Descartes $x^3 + y^3 = 6xy$ at the point $P(3, 3)$.
- (c) At what point in the first quadrant is the tangent line horizontal?

Example 23

- (a) Find y' if $\sin(x + y) = y^2 \cos x$.
- (b) Find y'' if $x^4 + y^4 = 16$.

3.6 The Implicit Differentiations

Derivatives of Inverse Trigonometric Functions

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx} (\cot^{-1} x) = -\frac{1}{1+x^2}$$

Example 24

Differentiate each of the following functions

a. $f(x) = \tan^{-1} \sqrt{x}$ **b.** $g(x) = \sqrt{1-x^2} \arccos x$ **c.** $h(x) = \arctan(x^2 - x)$

Sketch the proof

Find y' if $y = \sin^{-1} x$. Recall the definition of the arcsine function:

$$y = \sin^{-1} x \iff \sin y = x, \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}.$$

Differentiating the equation $\sin y = x$ with respect to x , we obtain

$$\cos y \frac{dy}{dx} = 1 \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{\cos y}$$

Now $\cos y \geq 0$, since $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, so

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

Therefore

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - x^2}}$$

3.6 The Implicit Differentiations

Logarithmic Differentiation

Logarithmic Differentiation

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the Laws of Logarithms to simplify.
2. Differentiate implicitly with respect to x .
3. Solve the resulting equation for y' .

Example 25

Differentiate

a. $y = \frac{x^{3/4} \sqrt{x^2 + 1}}{(3x + 2)^5}$

b. $y = x^{\sqrt{x}}$

3.7 Related Rates and Applications

Certain practical problems involve a functional relationship

$$y = f(x) \quad \text{or} \quad F(x, y) = 0 \quad (7)$$

where $x = x(t)$ and $y = y(t)$. Using implicit differentiation we can relate the rate of change dy/dt to the rate dx/dt . This is so-called **related rate problems**.

Example 26

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50 cm?

Example 27 (Learning ladder problem)

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?

Example 28 (The water level in a cone-shaped tank)

A water tank has the shape of an inverted circular cone with base radius 2 m and height 4 m. If water is being pumped into the tank at a rate of $2 \text{ m}^3/\text{min}$, find the rate at which the water level is rising when the water is 3 m deep.

Example 29 (Angle of elevation)

Every day, a flight to Los Angeles flies directly over my home at a constant altitude of 4 mi. If I assume that the plane is flying at a constant speed of 400 mi/h, at what rate is the angle of elevation of my line of sight changing with respect to time when the horizontal distance between the approaching plane and my location is exactly 3 mi?

Example 30

Car A is traveling west at 50 mi/h and car B is traveling north at 60 mi/h. Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3 mi and car B is 0.4 mi from the intersection?

Example 31

A man walks along a straight path at a speed of 4 ft/s. A searchlight is located on the ground 20 ft from the path and is kept focused on the man. At what rate is the searchlight rotating when the man is 15 ft from the point on the path closest to the searchlight?

3.8 Linear Approximation and Differentials

Linear Approximation

- The approximation

$$f(x) \approx f(a) + f'(a)(x - a) \quad (8)$$

is called the **linear approximation** or **tangent line approximation** of f at a .

- The linear function whose graph is this tangent line, that is,

$$L(x) = f(a) + f'(a)(x - a) \quad (9)$$

is called the **linearization** of f at a .

Example 32

- (a) Find the linearization of the function $f(x) = \sqrt{x}$ at $a = 4$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$. Are these approximations overestimates or underestimates?
- (b) For what values of x is the linear approximation in (a) accurate to within 0.5? What about accuracy to within 0.1?

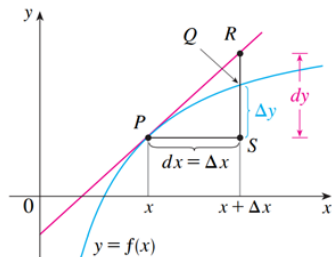
3.8 Linear Approximation and Differentials

Differentials

The **differential** of a differentiable function $y = f(x)$ denoted dy , is defined by

$$dy = f'(x)dx \quad (10)$$

where dx is an independent variable.



Let $P(x, f(x))$ and $Q(x + \Delta x, f(x + \Delta x))$.
Then change in y

$$\Delta y = f(x + \Delta x) - f(x) \approx dy \quad (11)$$

when $dx = \Delta x$ is small.

Application of Differentials: Error Propagation

Given a functional relationship $y = f(x)$ and assume that

- x is the **measured value** of a variable
- Δx is the **error in measurement**

Then the corresponding error in y which is called the **propagated error** at x

$$\Delta y = f(x + \Delta x) - f(x) \approx dy \quad (12)$$

The **relative error** is

$$\frac{\Delta y}{y} \approx \frac{dy}{y} \quad (13)$$

The errors can also be expressed as **percentage error** $100 \frac{\Delta y}{y}$ (%).

Example 33

The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using this value of the radius to compute the volume of the sphere?

Example 34 (Exercise 29, p. 212)

At an certain factory, the daily output is

$$Q(L) = 60,000L^{1/3}$$

units, where L denotes the size of the labor force measured in worker-hours. Currently 1,000 worker-hours of labor are used each day. Estimate the effect on output that will be produced if the labor force is cut to 940 worker-hours.